

```
In [1]: import numpy as np
import pandas as pd
```

```
In [2]: df=pd.read_csv('Iris.csv')
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   sepal.length    150 non-null    float64
1   sepal.width     150 non-null    float64
2   petal.length    150 non-null    float64
3   petal.width     150 non-null    float64
4   variety         150 non-null    str
dtypes: float64(4), str(1)
memory usage: 7.2 KB
```

```
In [3]: df.variety.value_counts()
```

```
Out[3]: variety
Setosa      50
Versicolor  50
Virginica   50
Name: count, dtype: int64
```

```
In [4]: df.head()
```

```
Out[4]:
```

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa

```
In [10]: import numpy as np
features=df.iloc[:, :-1].to_numpy(dtype=float)
label=df.iloc[:, 4].astype(str).to_numpy(dtype=object)
```

```
In [6]: from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
```

```
In [11]: xtrain,xtest,ytrain,ytest=train_test_split(features,label,test_size=.2,random_st
model_KNN=KNeighborsClassifier(n_neighbors=5)
model_KNN.fit(xtrain,ytrain)
```

Out[11]:

▼

KNeighborsClassifier

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► Parameters

► Fitted attributes



```
In [12]: print(model_KNN.score(xtrain,ytrain))
print(model_KNN.score(xtest,ytest))
```

0.95

0.9666666666666667

```
In [13]: from sklearn.metrics import confusion_matrix
confusion_matrix(label,model_KNN.predict(features))
```

```
Out[13]: array([[50,  0,  0],
               [ 0, 45,  5],
               [ 0,  2, 48]])
```

```
In [14]: from sklearn.metrics import classification_report
print(classification_report(label,model_KNN.predict(features)))
```

	precision	recall	f1-score	support
Setosa	1.00	1.00	1.00	50
Versicolor	0.96	0.90	0.93	50
Virginica	0.91	0.96	0.93	50
accuracy			0.95	150
macro avg	0.95	0.95	0.95	150
weighted avg	0.95	0.95	0.95	150

In []: